

RAHUL PODDAR

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SUMMARY

Senior Backend Software Engineer with 6+ years of experience building data-intensive backend systems using Java and Spring Boot. Proven track record delivering REST APIs, high-volume data ingestion pipelines, RBAC authorization, and database performance optimizations in production healthcare platforms. Hands-on experience designing Kafka-based event-driven systems and integrating AI/LLM-driven features into backend services with structured output parsing.

TECHNICAL SKILLS

Languages & Databases:	Java, Oracle (SQL, PL/SQL), PostgreSQL, MySQL
Backend & APIs:	Spring Boot, Spring MVC, JPA/Hibernate, MyBatis, RESTful APIs
Architecture & Patterns:	Distributed Systems, Event-Driven Architecture, Caching (Caffeine, Redis), Role-Based Access Control
Servers & Build:	Kafka, Docker, Maven, Git, Apache Tomcat, Oracle WebLogic, Zipkin (Distributed Tracing)
Testing & Quality:	JUnit, Mockito, SonarQube, JaCoCo
Frontend:	Thymeleaf, JavaScript, jQuery, AngularJS, HTML, CSS, AG Grid
AI & LLMs:	OpenAI API, Prompt Engineering, LLM Integration, Structured Output Parsing

WORK EXPERIENCE

Smart Source Technologies Inc

Albany, NY

Senior Software Engineer

Feb 2021-Present

Project- IMAPS (Information Management Application for Partner Services)

- Owned backend development of IMAPS, a centralized investigation platform modernizing a legacy public-health MIS system, used for managing workflows across multiple disease programs (HIV, Syphilis, Chlamydia, Gonorrhea, Mpox).
- Designed Spring Boot REST APIs and service-layer workflows, integrating complex Oracle PL/SQL stored procedures, and optimizing queries for high-volume investigation workflows, while enforcing secure access control using Spring Security RBAC.
- Diagnosed investigation workflow latency using Zipkin-based tracing with correlation IDs and optimized it by eliminating redundant database writes, adding proper indexing, and implementing pagination — reducing endpoint latency from ~500ms to ~20–40ms.

Project- NYEHMS (New York Electronic HIV Monitoring System) Web Application (Continued support)

- Improved reliability of ingestion workflows during production transitions for high-volume healthcare data feeds.
- Optimized ingestion performance by redesigning batch workflows, reducing bottlenecks, and improving system throughput.

Health Research Incorporated

Albany, NY

Application Development Specialist III (May 2019-Feb 2021) | Software Developer/Engineer (May 2018 – May 2019)

Project- NYEHMS (New York Electronic HIV Monitoring System) Web Application

- Contributed to backend development of a statewide HIV monitoring platform ingesting high-volume clinical data feeds (CCD XML, HL7, and death records) from regional healthcare partners.
- Built data ingestion workflows to parse HL7, CCD XML, CSV, and text-based healthcare documents, processing thousands of records daily into normalized patient and laboratory datasets using Java and Oracle PL/SQL.
- Designed resilient error-recovery workflows to handle and reprocess failed records during batch processing.

Project – Environmental Public Health Tracking Application

- Developed backend business logic and RESTful services using Java, Spring Boot, and Hibernate for public-health applications.
- Integrated backend services with AngularJS-based UI components and supported enterprise deployments on Oracle WebLogic Server.

DISTRIBUTED SYSTEMS PROJECT

Event-Driven Case Management System (Kafka Microservices)

Technologies: Java, Spring Boot, Kafka (Redpanda), PostgreSQL, Docker, Flyway

- Built an event-driven investigation platform using Kafka to model patient resolution, case lifecycle, and lab ingestion workflows.
- Designed loosely coupled microservices using Kafka topics, partitioning, and schema contracts to ensure ordering and scalability.
- Implemented idempotent consumer processing and transactional guarantees ensuring correctness under Kafka's at-least-once delivery.
- Engineered **fault-tolerant ingestion pipelines** with retries, DLQ handling, and persistent error tracking for observability and replay.

AI PROJECTS

AI API Assistant | Technologies: Java, Spring Boot, OpenAI API

- Built an AI-assisted backend system that generates structured API design suggestions (endpoints, DTOs, validation, error handling) from high-level requirements.
- Designed a prompt engineering and parsing pipeline to convert non-deterministic LLM output into strongly typed Java DTOs with enforced schema validation.

AI Log Analyzer | Technologies: Java, Spring Boot, OpenAI API

- Developed a backend service that analyzes application logs and generates summaries, root cause insights, and suggested fixes using LLMs.
- Implemented preprocessing and structured parsing to extract relevant log context and map AI output into typed fields for reliable downstream usage.

EDUCATION

Executive Master of Business Administration, University of the Cumberlands

May 2021-Dec 2022

Master of Science (MS), Computer Science, University at Albany SUNY

Aug 2016-May 2018

Bachelor of Engineering, Computer Science, Madhav Institute of Technology & Science

Aug 2011-May 2015